

Surviving A HARD DISK CRASH



Crash! Bang! Kaboom! There goes your hard disk. But here's how you can recover your data and live to tell the story

edge, the platter surface could get scratched resulting in data loss. Most recent hard drives come equipped with an automatic platter parking mechanism, so chances of this happening are rare if the drive isn't powered on.

Electronics Failure: A power surge can easily kill the hard drive's logic board, or even the small servo motors used inside. The hard drive head uses an electromagnetic actuator to move between positions. It is possible that the actuator's circuit might get damaged, rendering the read mechanism useless.

Media Errors/Malformed Data: Due to manufacturing defects, or just plain rugged usage, a certain part of the platter can get physically damaged beyond recovery, rendering that part of the platter useless. When this happens, a disk utility generally marks all sectors in this

Mechanical Faults:

If you bump your hard disk around, chances are it will give up on you. While transporting your hard disk, if the hard drive's read/write head isn't parked properly at the platter's non-data storage

You're at the fag end of your financial year. You've been working ever so diligently on your year-end presentation. You switch on your PC for the final bunch of printouts and you're faced with a screen saying 'disk boot failure'. Welcome to your worst nightmare—the day your hard disk crashes!

What causes a hard disk crash?

Like any electrical appliance or product, a hard disk is prone to malfunctions that could lead to premature death. Your hard disk may be working just fine, but suddenly one day your computer refuses to boot up. Let's take a look first at how a hard disk crash could occur.

Diagnosing the Problem	
SYMPTOM	AILMENT
BIOS fails to detect the hard drive	Power failure, disconnected cables, drive electronics failure
Random crashes; error messages; drive detected, but unusable	Media errors, read/write errors due to other reasons, unmarked bad sectors
Sudden data loss; missing partitions; boot errors	Media corrupt, demagnetised
Drive makes noise/click sounds	Virus attack on boot sector, file allocation table corrupt, partition table corrupt/modified, head actuator mechanism error, head parking mechanism damaged